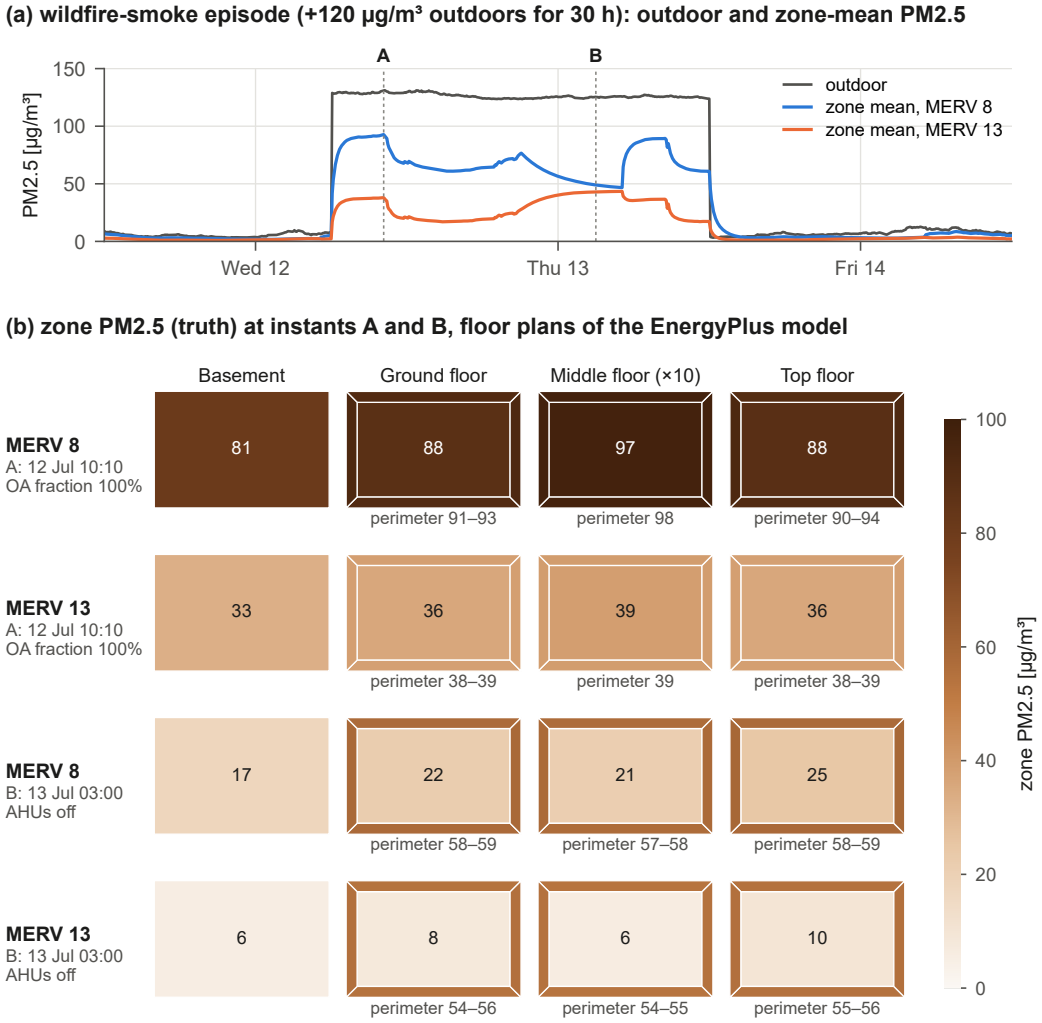

(a) wildfire-smoke episode (+120 µg/m³ outdoors for 30 h): outdoor and zone-mean PM2.5
A
B
150
100
50
0
PM2.5 [µg/m³]
outdoor
zone mean, MERV 8
zone mean, MERV 13
Wed 12
Thu 13
Fri 14
(b) zone PM2.5 (truth) at instants A and B, floor plans of the EnergyPlus model
Basement
Ground floor
Middle floor (×10)
Top floor
MERV 8
A: 12 Jul 10:10
OA fraction 100%
81
88
97
88
perimeter 91–93
perimeter 98
perimeter 90–94
MERV 13
A: 12 Jul 10:10
OA fraction 100%
33
36
39
36
perimeter 38–39
perimeter 39
perimeter 38–39
MERV 8
B: 13 Jul 03:00
AHUs off
17
22
21
25
perimeter 58–59
perimeter 57–58
perimeter 58–59
MERV 13
B: 13 Jul 03:00
AHUs off
6
8
6
10
perimeter 54–56
perimeter 54–55
perimeter 55–56
100
80
60
40
20
0
zone PM2.5 [µg/m³]